

Finite Polar-Space Permutation Actions

Unitary, symplectic, and orthogonal σ -MGFs

Proof and computational verification companion

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Abstract

This note specializes the exact permutation σ -MGF to isotropic and singular point actions of finite unitary, symplectic, and orthogonal groups. Witt extension proves the stable three-orbit classification of pairs. The companion marimo notebook constructs $Sp_4(2)$, $O_4^+(2)$, and $U_4(2)$ as finite-field matrix groups, induces their point actions, and independently compares direct orbit counts, cycle products, fixed-point reconstruction, and matrix determinants. Every tested coefficient agrees exactly.

1 Exact formula for a polar action

Let a finite classical group G act on a finite set S of isotropic or singular one-spaces. For $g \in G$, let $c_d(g; S)$ denote the number of d -cycles of its action on S . The permutation-matrix eigenvalues on a d -cycle are the d th roots of unity, so

$$\mathcal{S}_{G,S}(t_\bullet) = \frac{1}{|G|} \sum_{g \in G} \prod_{i, d \geq 1} (1 - t_i^d)^{-c_d(g; S)}. \quad (1)$$

Moreover,

$$\text{Fix}_S(g^r) = \sum_{d|r} d c_d(g; S), \quad c_d(g; S) = \frac{1}{d} \sum_{r|d} \mu(d/r) \text{Fix}_S(g^r). \quad (2)$$

Thus one may compute (1) purely from the number of isotropic or singular eigenlines of every power g^r . Equivalently, for a parabolic point stabilizer P , this fixed-point count is $(\text{Ind}_P^G \mathbf{1})(g^r)$.

For $\tau = (\tau_1, \dots, \tau_s)$, expanding the determinants gives

$$[m_\tau] \mathcal{S}_{G,S} = \dim \left(\bigotimes_{b=1}^s \text{Sym}^{\tau_b} \mathbb{C}[S] \right)^G. \quad (3)$$

The tensor product has a basis consisting of tuples of multisets of points. Because G permutes this basis, (3) is exactly the number of G -orbits on such tuples. This orbit count is the independent direct computation used below.

2 Why the degree-two coefficient is three

For two isotropic or singular points in a nondegenerate polar space of sufficient Witt rank, there are three possible geometric types:

1. the two points coincide;
2. they are distinct and perpendicular;
3. they are distinct and nonperpendicular.

Each relation is preserved by every isometry. Conversely, an isometry between either pair of generated subspaces extends to the ambient space by Witt's extension theorem. Hence each type is one orbit. This applies to both ordered pairs and unordered 2-multisets, proving

$$\boxed{[m_{(2)}]\mathcal{S} = [m_{(1,1)}]\mathcal{S} = 3.} \quad (4)$$

It follows that the stable expansion begins

$$\mathcal{S} = 1 + m_{(1)} + 3m_{(2)} + 3m_{(1,1)} + O(\deg 3).$$

The extra perpendicularity relation already distinguishes these limits from the Poisson/symmetric-group series in degree two.

Proposition (coefficientwise stability within a fixed polar tower). Fix the field, one classical type, and a total degree D . For each tuple of multisets of singular or isotropic points of total degree D , retain the labelled span together with the restricted form and the multiplicities of the labelled one-spaces; in characteristic-two orthogonal cases retain the restricted quadratic form rather than only its polar bilinear form. Two configurations in the same ambient nondegenerate polar space are conjugate precisely when these labelled formed spaces are isometric, because an isometry of their spans extends by Witt's lemma.

Only finitely many labelled formed spaces of dimension at most D exist over the fixed finite field. Let $m_0(D)$ be the maximum, over the types that occur somewhere in the chosen tower, of the least Witt rank in which the type embeds. Every coefficient of total degree D is therefore constant for Witt rank at least $m_0(D)$. Orthogonal plus, minus, and odd-dimensional towers are treated separately. This proves existence of a stable threshold but does not claim that the displayed bound is sharp. The degree-two value three was proved directly above and does not depend on this proposition.

3 The three exact matrix constructions

All arithmetic in the program is exact integer arithmetic.

$Sp_4(2)$. We enumerate $GL_4(2)$ and retain exactly the matrices preserving

$$B(x, y) = x_1y_2 + x_2y_1 + x_3y_4 + x_4y_3.$$

The resulting group has order 720. In characteristic two every one-space is isotropic, giving a permutation action on the 15 nonzero vectors (equally, the 15 projective points).

$O_4^+(2)$. We retain the matrices preserving the hyperbolic quadratic form

$$Q(x) = x_1x_2 + x_3x_4.$$

The resulting group has order 72 and acts on its 9 nonzero singular vectors.

$U_4(2)$. We implement $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$ and the Hermitian form

$$h(x, y) = \sum_{j=1}^4 \bar{x}_j y_j, \quad \bar{z} = z^2.$$

Every ordered orthonormal basis is enumerated recursively, producing exactly

$$|U_4(2)| = 2^6 \prod_{i=1}^4 (2^i - (-1)^i) = 77,760$$

matrices. Normalizing each nonzero vector by its first nonzero coordinate gives 45 isotropic projective points. The scalar center has order three, so the faithful induced permutation image has order 25,920.

4 Verification strategy and results

For each action, the notebook:

1. constructs the full finite-field matrix group and induced permutation;
2. directly enumerates orbits on ordered point pairs and 2-multisets;
3. extracts the same coefficients from the cycle product in (1);
4. reconstructs cycle counts from fixed points of powers using (2); and
5. numerically compares direct permutation-matrix determinants with the cycle products at $(t_1, t_2) = (0.07, 0.11)$.

The regression module also compares a degree-three labelled-formed-space orbit count in two consecutive ranks of each implemented tower. This tests the restricted-form encoding; it is not used as a proof of the proposition.

Action	τ	direct orbits	cycle formula	claimed
$Sp_4(2)$ on 15 isotropic points	(2)	3	3	3
	(1, 1)	3	3	3
$O_4^+(2)$ on 9 singular points	(2)	3	3	3
	(1, 1)	3	3	3
$U_4(2)$ on 45 isotropic points	(2)	3	3	3
	(1, 1)	3	3	3

Every equality in the table is exact. All cycle counts in the tested $Sp_4(2)$ and projective $U_4(2)$ actions are also recovered exactly from fixed points of powers. For the symplectic action, the direct determinant average and cycle-product average both equal 1.304867127368, with zero difference at the displayed floating-point precision.

5 Representations requiring additional choices

The Weil-representation paragraph is governed by the general identity

$$\det(I - t\rho(g))^{-1} = \exp \left(\sum_{r \geq 1} \frac{\chi_\rho(g^r)}{r} t^r \right),$$

but it does not specify one numerical representation. A reproducible matrix test requires a choice of q and n , an additive character, a linearization, and whether ρ is the full Weil representation or a constituent. Even characteristic requires a different construction. Accordingly, no arbitrary choice was silently inserted into the verification.

Likewise, the exceptional/Suzuki/Ree formula is an exact template for a specified pair $(G(q), P(q))$, not a single representation. A numerical test requires the concrete group, field size, and parabolic action. Once supplied, the same fixed-point, cycle, determinant, and orbit pipeline applies without change.

Conclusion. The three classical polar families realize exactly the predicted three pair types in small but stable-rank examples. The agreement of full matrix-group enumeration, direct configuration orbits, cycle extraction, Möbius reconstruction, and determinants validates both the formula and the proposed verification method.